

UPDF Family Welfare Portal

Architecture, installation and operations guide

1. Overview

A welfare support portal for Uganda People's Defence Force families. Family members submit support requests with documents; officers and admins triage, approve and record aid disbursements in the ledger. Built with TanStack Start (React 19 + Vite 7), Tailwind v4, shadcn/ui and Lovable Cloud (Supabase) for authentication, database, storage and server functions.

2. Roles & access control

Role	Capabilities
family	Create and view own support requests; upload documents to own folder.
officer	View all profiles, requests and documents; update request status.
admin	Everything officers can do, plus grant/revoke roles and view audit log.

Every public table has Row-Level Security enabled. Privilege checks go through the SECURITY DEFINER function **has_role(user, role)**. A restrictive RLS rule blocks any user from inserting or updating their own row in **user_roles**, so self-elevation is impossible even if another policy is misconfigured.

3. Data model

- **profiles** – full_name, service_number, linked to auth user.
- **user_roles** – role assignments (enum: family / officer / admin).
- **support_requests** – title, type, urgency, status, details.
- **request_documents** – uploaded files stored in the support-documents bucket.
- **aid_ledger** – recipient, region, aid type, amount, status, disbursed_at.
- **role_change_audit** – append-only via SECURITY DEFINER trigger.

4. Run locally

Install **Bun 1.1+** (or Node 20 + npm) and Git, then:

```
git clone <your-repo-url>
cd <repo>
bun install
cp .env.example .env # then edit values
bun run dev # http://localhost:8080
```

Environment variables (.env at repo root):

```
VITE_SUPABASE_URL=https://<project>.supabase.co
VITE_SUPABASE_PUBLISHABLE_KEY=<publishable key>
VITE_SUPABASE_PROJECT_ID=<project id>

# Server-only – never commit or expose in client code
SUPABASE_URL=https://<project>.supabase.co
SUPABASE_PUBLISHABLE_KEY=<publishable key>
```

```
SUPABASE_SERVICE_ROLE_KEY=<service role key>
```

The publishable/anon key is safe in the browser. The service role key bypasses all RLS — keep it server-side only. On Lovable Cloud it is injected automatically; for local dev paste it from your Supabase project's API settings.

5. Push to GitHub

Inside Lovable: top-right + menu → **GitHub** → **Connect project**, authorise the Lovable GitHub App, choose an account or organisation, then click **Create Repository**. Sync is two-way — edits in Lovable push automatically, and pushes to GitHub appear back in Lovable.

Clone the new repo locally to develop in your own editor:

```
git clone git@github.com:<you>/<repo>.git
cd <repo>
bun install
bun run dev
```

6. Deploy

- **Lovable** – click **Publish** in the top-right; configure a custom domain in project settings.
- **Cloudflare Workers / Pages** – TanStack Start targets edge runtimes:

```
bun run build
# deploy .output/ with wrangler, Cloudflare Pages, etc.
```

Set the same environment variables on the host. Database migrations live in **supabase/migrations/** and can be applied with **supabase db push**.

7. Project layout

```
src/
routes/ file-based routes (TanStack Router)
_authenticated/ protected subtree (auto-gated)
api/ server route handlers (webhooks, public APIs)
components/ shared UI + shadcn
hooks/ use-auth, etc.
integrations/
supabase/ auto-generated clients (do NOT edit)
lib/ shared utilities, *.functions.ts server functions
supabase/migrations/ SQL migrations
docs/ this PDF
```

Server logic lives in ***.functions.ts** via `createServerFn`. Webhooks and public APIs go under **src/routes/api/public/**. Never import **client.server.ts** at module scope from client-reachable files.

8. Security checklist

- RLS enabled on every public table.
- Restrictive policy blocks self privilege escalation on `user_roles`.
- `role_change_audit` is append-only via SECURITY DEFINER trigger; restrictive policies block direct writes.
- support-documents bucket is private and scoped by `auth.uid()` folder, including UPDATE.

- Service role key is never shipped to the browser.
- Authenticated routes live under `src/routes/_authenticated/` and are gated by the managed layout.

9. Support

For questions about Lovable hosting, GitHub sync or Cloud, see docs.lovable.dev. For Supabase-specific questions see supabase.com/docs.